

MANCHESTER-BOSTON, NH, USA

WMO: 743945

Lat: 42.930N

Lon: 71.436W

Elev: 221

StdP: 14.58

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14710

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.8	7.0	-13.1	2.7	6.4	-8.3	3.5	11.8	23.0	19.8	20.6	22.4	8.0	330	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	91.0	71.6	88.4	70.4	85.6	69.0	75.1	85.0	73.5	82.5	72.2	81.0	9.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.2	120.3	79.4	70.9	114.8	78.2	69.6	109.5	77.2	38.6	85.0	37.2	82.8	36.0	80.3	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.2	18.0	16.0		DB	-4.6	96.2	5.4	3.1	-8.5	98.5	-11.6	100.3	-14.6	102.1	-18.5	104.4
				WB	-5.3	78.1	4.9	2.2	-8.9	79.8	-11.8	81.1	-14.5	82.3	-18.1	84.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	25.4	28.2	36.1	48.1	58.9	68.0	73.8	72.0	64.4	52.3	41.4	31.5
	DBStd	18.57	10.56	9.59	9.26	8.33	7.75	7.17	5.47	5.30	6.95	7.57	8.52	8.82
	HDD50	2880	763	613	441	131	10	0	0	0	1	62	282	577
	HDD65	6191	1227	1031	896	512	225	49	4	7	94	399	708	1039
	CDD50	2922	1	1	11	73	285	541	736	681	433	134	24	2
	CDD65	761	0	0	1	4	35	140	276	223	77	5	0	0
	CDH74	6869	0	0	13	85	461	1311	2507	1840	597	53	2	0
	CDH80	2363	0	0	3	24	153	465	946	595	171	6	0	0
Wind	WSAvg	6.6	7.0	7.3	7.8	7.7	6.9	6.3	5.9	5.5	5.7	6.1	6.4	6.6
Precipitation	PrecAvg	43.9	3.2	3.0	3.8	3.6	3.6	3.8	3.4	3.7	3.7	4.2	3.8	4.1
	PrecMax	63.8	6.1	10.3	11.1	9.2	13.0	10.8	8.1	8.9	8.7	13.9	6.1	6.5
	PrecMin	30.2	1.2	1.3	0.7	0.7	0.7	0.9	1.3	0.3	1.3	0.5	1.4	1.2
	PrecStd	8.3	1.2	1.7	2.2	2.1	2.4	2.5	1.8	2.1	2.0	2.9	1.2	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.0	60.3	72.9	83.3	90.0	93.4	94.8	93.2	90.2	79.7	70.0	61.4
		MCWB	56.3	51.0	57.1	60.1	66.5	72.2	74.5	73.5	71.0	64.3	61.1	52.8
	2%	DB	50.3	51.9	62.5	74.6	84.3	89.8	91.1	89.6	84.8	74.1	64.0	53.7
		MCWB	44.8	43.9	51.0	56.9	64.3	70.8	72.5	71.4	69.1	62.7	56.8	48.8
	5%	DB	44.3	46.0	55.0	68.5	79.2	85.4	88.4	86.1	80.8	69.9	59.8	48.3
		MCWB	39.5	39.3	44.5	53.9	61.3	68.4	71.0	69.8	67.2	60.8	52.8	43.4
	10%	DB	39.0	41.3	49.8	63.2	73.9	81.6	85.5	83.1	76.7	65.1	54.8	44.3
		MCWB	34.7	35.8	41.6	50.2	58.9	66.1	69.7	68.7	65.2	57.1	48.8	39.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.0	54.4	59.2	64.6	70.4	76.1	77.6	76.6	74.5	69.1	63.4	56.0
		MCDB	59.0	58.7	70.4	74.7	84.5	88.8	90.2	87.8	82.3	73.9	67.0	59.7
	2%	WB	46.0	45.2	52.7	60.0	67.6	73.4	75.4	74.4	72.1	65.4	58.9	49.6
		MCDB	48.6	49.5	59.7	71.0	78.3	83.9	85.8	83.1	79.9	71.4	63.1	52.5
	5%	WB	39.6	39.8	46.4	56.0	64.7	71.6	73.8	72.9	70.1	61.8	54.0	44.1
		MCDB	43.7	45.1	52.7	65.6	75.0	81.3	83.3	81.2	77.1	68.2	57.6	47.6
	10%	WB	35.4	36.5	42.0	51.4	61.7	69.3	72.3	71.3	68.0	58.6	49.9	39.6
		MCDB	38.5	40.8	48.7	61.0	71.6	77.2	81.3	79.6	74.6	64.1	54.7	43.4
Mean Daily Temperature Range	5% DB	MDBR	15.4	16.8	17.0	19.8	20.3	19.2	19.0	18.9	19.0	18.2	15.7	14.3
		MCDBR	20.3	22.1	25.5	29.7	28.9	24.7	22.6	22.3	23.5	25.1	21.8	20.3
		MCWBR	17.1	16.2	16.0	15.5	13.2	10.3	8.4	8.6	10.7	14.5	15.8	16.5
	5% WB	MCDBR	19.4	20.8	23.7	26.0	24.7	21.1	19.4	18.5	18.6	21.0	18.7	19.3
		MCWBR	17.2	16.3	16.8	16.4	13.0	10.2	8.9	8.8	10.6	14.8	15.8	17.1
Clear-Sky Solar Irradiance	taub		0.284	0.295	0.321	0.360	0.396	0.437	0.459	0.426	0.372	0.332	0.308	0.288
	taud		2.452	2.431	2.451	2.395	2.332	2.241	2.192	2.267	2.401	2.523	2.548	2.533
	Ebn at Noon		274	289	293	288	279	266	259	264	272	272	263	263
	Edh at Noon		23	28	31	36	39	43	45	41	33	26	22	20
All-Sky Solar Radiation	RadAvg		555	829	1156	1459	1611	1742	1836	1659	1342	865	593	448
	RadStd		35	64	76	159	193	166	117	85	89	81	53	48

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+164	N/A
Regional (79 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+162	+65

Nomenclature: See separate page